



Manifold for T1D – HIRN Cross-Consortium Integration

Cross-consortium multi-omic causal integration across CBDS · CTAR · CHIB · HIPPI

HIRN was established explicitly on the recognition that T1D restoration is a cross-modality integration problem. Its four consortia — **CBDS** (β-cell death and survival), **CTAR** (targeting and regeneration), **CHIB** (human-islet biomimetics and bioengineering), and **HIPP** (human-islet phenotyping) — each produce rich, orthogonal signal on the same biological substrate. The field-level analytic gap is the integration itself: the directed structure connecting death cascades, regenerative targets, engineered substrates, and phenotypic profiles into a single causal graph. **Manifold** configures three engine capabilities to the HIRN substrate: causal discovery on cross-consortium multi-omic data, criticality estimators (persistent homology on HIPP manifolds, HTE learners on CTAR data, change-point on CHIB substrates, survival models on CBDS markers) feeding a composite fragility score (ΩF), and Pearl do-calculus with interdiction for principled cross-consortium target prioritization.

HIRN's constituent consortia are individually world-class. The field-level question is whether their combined output can be integrated on a single causal substrate to produce translational leverage that exceeds the sum of consortium analyses — a problem of cross-consortium causal inference.

ΩF PILLAR MAPPING – HIRN CROSS-CONSORTIUM PHYSIOLOGY

ΩF PILLAR	HIRN READING
I — Interaction	Cross-consortium signal interactions: death-cascade × regenerative-target × engineered-substrate × phenotype coupling
R — Reserve	Baseline and engineered β-cell functional mass across HIPP and CHIB substrates
J — Junction	Immune-metabolic and death-regeneration cascade junctions; cross-consortium transfer-entropy structure
C — Cascade	CBDS death cascades (ER stress, apoptosis, autophagy) interacting with CTAR regenerative cascades (DYRK1A, mTOR, FOXO1)
T — Temporal	Hazard of β-cell loss, graft compromise, or regenerative response loss across the integrated pipeline

PROPOSED PARTNERSHIP

Partnership with HIRN coordinating-center leadership and co-sponsoring consortium principal investigators is sought for:

1. Cross-consortium retrospective data access under the HIRN coordinating-center Data Use Agreement framework.
2. Co-authorship on the cross-consortium causal-integration methodological manuscript, positioned as a HIRN-affiliated deliverable (twelve to sixteen weeks from data access).
3. Co-application to NIH NIDDK HIRN expansion, Breakthrough T1D Cures, or the Helmsley Charitable Trust for the Phase 2 arm extending integration across additional consortia.

Apex Analytica contributes the causal-inference engine, the ΩF framework, and the engineering team. Full brief: M-T1D-02-HIRN .